# Assigned to your Choice? A short Note on Assignment of Researchers to Workspaces in a Research Department*

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**Abstract**

In 2009, while in our roles as Researcher Representatives (Reps) for the Economics Department at the EUI, one of our tasks was to assign researchers from 2nd, 3rd and 4th year to available workspaces. We decided to implement an algorithm which is commonly used in, among others, the problem of assigning students to schools, and which has its origins on the matching problem developments for which A. Roth and L. Shapley received the 2012 Economics Prize in Memory of A. Nobel. We present a short summary of the procedure and algorithm carried out to assign researchers to workspaces. We then use Monte Carlo simulations in order to analyze its behavior and to compare it with alternative algorithms commonly used for similar problems. We find that the implemented algorithm has interesting properties in terms of final allocations.

## 1 Introduction: Assignment and Matching Problems

Many real situations require the assignment or matching of elements / individuals / resources to tasks or other elements: students need to be assigned to colleges that don't have enough spaces to accept all the applicants; individuals match each other in order to form couples and marriages; donated organs need to be matched with feasible patients; firms assign transport routes to cargo vehicles, etc. The way in which these problems can be solved depends crucially on, among others, two features: (i) whether assignments or matching need to be one-to-one, meaning that each element can only be matched to one other element, and (ii) whether each element has a strict preference rank for all other elements to which it can be assigned or matched.

The search for efficient algorithms to solve these problems has given way to important contributions in the last 50 years. In particular, ss released on November 8, 2012 in Nobelprize.org, the 2012 Economics Prize was awarded to Lloyd Shapley and Alvin Roth

*Regorously speaking, the prize is called The 2012 Economics Prize in Memory of A. Nobel.

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> "... *for the theory of stable allocations and the practice of market design. Examples include the assignment of new doctors to hospitals, students to schools, and human organs for transplant to recipients. Lloyd Shapley made the early theoretical contributions, which were unexpectedly adopted two decades later when Alvin Roth investigated the market for U.S. doctors. His findings generated further analytical developments, as well as practical design of market institutions.*"

Broadly speaking, an *assignment problem* is one in which there are a number of tasks that need to be performed, and there is a number of agents that can perform these tasks. The cost of carrying a certain task varies across agents. The problem then boils down to assigning exactly one agent to each task in a way that the total cost of the assignment is minimized. A *matching problem* deals with finding a matching (map) between two set of elements (or individuals) when the elements of each set have elicited *preferences* for all the elements (individuals) of the opposite set. The matching is said to be *unstable* if, for a given individual $i$ from the set A and individual $j$ from the set B who have not been matched together, $i$ prefers $j$ over her current match *and* $j$ prefers $i$ over her current match.

In a first seminal paper, Gale and Shapley (1962) describe a procedure for assigning applicants to colleges that takes into account their ranked preferences and would be satisfactory for both applicants and colleges. In the special case in which the number of colleges equals the number of applicants, the result is a stable matching—one in which no element of the first matched set prefers an element of the second matched set that also prefers the first element. The Gale-Shapley algorithm for producing such optimal pairings is now often expressed in terms of matching men and women to create stable marriages. In a second seminal paper, Roth (1982)

## 2 Assignment of Researchers to Workspaces

In the Economics department at the European University Institute (EUI), Ph.D. students in the 2nd year and above get a workspace assigned to them at the beginning of each academic year. The assignment is valid for one year, after which the process is carried out again. In this way, the new 2nd year students (who were in 1st year one year before) are included, and those students who finished their stay at VSP during such year are dropped from the list.

In 2009, Workspaces in Villa San Paolo (VSP) and Villa San Paolino (VSPino), the two buildings of the economics department, are distributed asymmetrically across 13 office rooms according to table 1. Rooms are of different sizes and have different characteristics, including different number of workspaces, as well as different light conditions, noise, etc. This implies that students might be more inclined to work in certain offices than in others, which makes the allocation process non-trivial. The number of researchers during such year was 66 (this includes 2nd, 3rd, 4thyear researchers and visiting researchers at the department during the period)..

Table 1: Distribution of Workspaces during 2009

| Room id | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| # of workspaces | 3 | 4 | 4 | 8 | 6 | 5 | 7 | 5 | 8 | 4 | 4 | 2 | 6 |

The allocation process then has some parallels with the matching problems described above, although with an important difference: in the workspace allocation problem (WAP), the preferences revelation is one-sided. This means that workspaces, which can be thought as the other party in the matching problem, do not have to reveal preferences for students.

In addition, there are no "priorities" defined for different groups of students[1]; in other words, there were no workspaces or rooms pre-allocated to students with certain characteristics, except physical disabilities. This implies that in case that a room is over-demanded (i.e. the demand for such room is higher than the available workspaces in it), no student has an advantage over others who ranked that room *in the same position* as them.

# 3 Three Possible Allocation Algorithms

There are different ways in which researchers can be assigned to a particular workspace. Among these, we are going to consider three alternative which can be thought as resembling a Boston Mechanism (BM), a Deferred Acceptance (DA) and a Random Dictator (RD); we will label them Algorithm 1, 2 and 3, respectively. In both the BM and the DA students are asked to reveal preferences for rooms, and then they are allocated according to different alternative algorithms; under the RD case, researchers are ranled randomly and are then allowed to pick a room according to their rank.

An important difference between BM / DA and RD is the way in which each algorithm deals with the excess demand, and the potential effects of this on outcomes if students "internalize" the characteristics of the algorithms when revealing a ranking of rooms. In particular, researchers that understand how the BM algorithm works might decide to behave strategically: if a researcher believes that her most preferred room is highly demanded, she might decide to rank first a room other than her true most preferred one. The reason is te following: when her most prefered room is highly demanded, meaning that there are more researches wanting that room than available workspaces in it, then the chances that she will be assigned to it decrease. Since in the BM all individuals are considered simultaneously for their first choice and then for their second choice, when a researcher gets "rejected" from their 1st ranked room, all the available workspaces in her second choice might have already been taken by other researchers that opted for that room as their first choice. Acting strategically by avoiding to rank first a room with high demand might give her a higher chance of being effectively assigned to a room she has among her preferences. Interestingly, Miralles (2008) finds that when there are no priorities, or when priorities are coarse and lotteries are needed to break ties, the BM has preferable ex-ante efficiency properties from an utilitarian perspective compared to DA. This is so even when individuals have correlated utilities and when there are "naive" truth-telling individuals; naive individuals, however, are better-off under the DA algorithm.

In 2009, the process was carried out according to Algorithm 1, while in 2012 the allocation followed Algorithm 3 (a resemblance to RD). More specifically, researchers in 2009 were presented with all the available rooms (13) and workspaces in each of them, and asked to fill in a formulaire similar to the one in table 2.

[1] In 2009, however, students had the chance to add some commets to their preference revelation about particular needs, which might indeed create some individual priorities if such need where "reasonable".

Table 2: Preference Form

| Name: Year: | |
|---|---|
| $1^{st}$ choice | Room no.: ....... |
| $2^{nd}$ choice | Room no.: ....... |
| $3^{rd}$ choice | Room no.: ....... |
| Comments: | |

Once all the preferences were collected, A1 was run *once* in order to determine the final allocation. Crucially, when the researchers where asked to rank the rooms, they *were not told* which algorithm was going to be used for the final allocation.

A mentioned before, rooms do not have "preferences" or priorities for researchers. Under the "no priorities" rule, all researchers belong to the same status or group. One way to think about this is as an assignment problem where the "tasks" are the rooms, and where the "cost" of being assigned to a given room increases with the preference rank of such room (assignment to a room that is not in the rank list would have a cost $c \geq c_{3rd}$, where $c_{3rd}$ is the cost of being assigned to your $3^{rd}$ revealed preference). This interpretation, however, is meaningful only if individuals do not behave strategically, i.e. if their revealed ranking correspond to their true preference ranking. can help to understand the results and the comparison using different measures.

Bellow we describe the basics of each algorithm. In the next section we will present some performance measures of the three using both the ranking as revealed by students in 2009 and random draws of rankings for each individual.

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**Algortihm 1**

1. Researchers are asked to rank three workspaces where they would like to work. They where also given the chance to provide some exceptional reason why they might need some particular workspace

2. For preference $k \in \{1, 2, 3\}$, starting with preference 1, do the following:

3. For room $i \in \{1, ..., 13\}$, if there are still available workspaces, researchers who had selected room $i$ as a $k^{th}$ choice, and have not been allocated to their higher choices, are considered.

   - If $\#\left(\text{researchers with room } i \text{ as } k^{th}\text{choice}\right) < \#\,(\text{available workspaces in room } i)$, then allocate researchers and delete those spaces from the available spaces list.
   - If $\#\,(\text{researchers with room } i \text{ as 1st choice}) > \#\,(\text{workspaces in room } i)$, allocate the available workspaces randomly among the researchers who selected room $i$ as $k^{th}$ choice.

4. Move to next room and go back to step 3.

5. Repeat steps 3-4 until all rooms are considered. Then move to preference $k + 1$ and go back to step 2.

6. After all preferences are considered, allocate *randomly* the remaining (if any) researchers that where "unlucky" (or naive) to available workspaces.

**Algortihm 2**

1. Researchers are asked to rank three workspaces where they would like to work. They where also given the chance to provide some exceptional reason why they might need some particular workspace

2. For room $i \in \{1, ..., 13\}$, **all researchers** who had selected room $i$ as a $1^{st}, 2^{d}$ or $3^{rd}$ choice are considered.

3. For researcher $j$ who has room $i$ as $k^{th}$ preference do the following:

   - If # (researchers with room $i$ within their preferences) < # (available workspaces in room $i$), then TEMPORALLY allocate room $i$ to researcher $j$'s $k^{th}$preference.
   - If # (researchers with room $i$ within their preferences) > # (workspaces in room $i$), assign researcher $j$ a random number between 0 and 1.

4. Repeat step 3 for all researchers. At the end, each researcher has either a room or a random number assigned to each of her 3 preferences.

5. For each room that is over-demanded (i.e. random numbers have been assigned), sort in descending order all random numbers given to researchers who selected that room.

6. For each researcher, starting with her 1st choice: If such choice has not been assigned a room yet, and if she has a random number $\neq 0$ assigned for that choice, check whether she is sufficiently high in the sorting. If so, TEMPORALLY assign the chosen room to her choice, and then set the random numbers (if any) for her lower preferences to 0. In that way, she is "freeing" spaces for other rooms, to be considered in this or next round.

7. Repeat steps 5-6 until the sorting of random numbers does not change between two consecutive rounds..

8. Allocate the remaining (if any) researchers to available workspaces randomly.

**Algortihm 3**

1. Denote by $TotR \equiv$ total number of researchers. Researchers are assigned a random integer $r \in \{1, ...TotR\}$, and ranked in ascending order according to it.

2. Allow researcher $r$, starting with $r = 1$, to select a room / workspace.

3. If $r < TotR$, then move to researcher $r + 1$ and go back to step 2.

# 4 Quantitative Experiment

In order to carry out the comparison between the three allocation algorithms, we restort to Monte Carlo simulations. Concretely, we perform two sets of simulations:

1. The first set uses the fixed preferences of 2009. We then run each algorithm $N = 2000$ times, which generates $N$ different final allocations.

2. The second set uses $S = 1000$ random preference draws, and then runs the algorithms N times for each draw. In order to have results from both a symmetric and an asymmetric distribution, we draw both from a uniform and a log-normal distribution.

Recall that for all three algorithms, for a *given* preference draw (either the "true" preferences from 2009 or a random draw), there is uncertainty in the final outcome if there exist excess demand for at least one room. Such uncertainty aroused due to the randomization in the presence of an over-demanded room, though the individuals involved in the randomization vary between algorithms.

Each simulation generates an allocation outcome. For draw $s \in \{1, ..., S\}$ (note that when only the 2009 preferences are used, we have $S = 1$) and run $n \in \{1, ..., N\}$, we define the proportion of researchers that get allocated to their $k^{th}$ choice as:

$$prop_{s,n}^{k^{th}} = \frac{1}{66} \sum_{i=1}^{66} 1 \cdot \left( \text{researcher } i \text{ gets } k^{th} \text{ choice} \right)_{s,n} \tag{1}$$

where $1 \cdot ()$ is the indicator function. Note that $prop_{s,n}^{k^{th}}$ is a number between 0 and 1.

We then compute moments from the simulated distribution of $prop_{s,n}^{k^{th}}$:

- **Sample Average proportion** of researchers who get their $1^{st}, 2^{nd}, 3^{rd}$ choices or get randomly assigned to residual workspaces:

$$\overline{prop^{k^{th}}} = \frac{1}{S} \frac{1}{N} \sum_{s=1}^{S} \sum_{n=1}^{N} prop_{s,n}^{k^{th}} \tag{2}$$

- **Mediam proportion** of the distribution:

$$med\left(prop_{s,n}^{k^{th}}\right) = \left\{ prop^{med} : \; p\left(prop_{s,n}^{k^{th}} < prop^{med}\right) = 0.5 \right\} \tag{3}$$

- **Sample Standard deviation** of the distribution:

$$\hat{\sigma}\left(prop_{s,n}^{k^{th}}\right) = \left( \frac{1}{S} \frac{1}{N} \sum_{s=1}^{S} \sum_{n=1}^{N} \left( prop_{s,n}^{k^{th}} - \overline{prop^{k^{th}}} \right)^2 \right)^{\frac{1}{2}} \tag{4}$$

## 4.1 Results: Ranking Revelations from 2009

Table 3 presents that first set of results for the three algorithms, using the preference revelation of 2009. Columns 3 to 5 present the values for $\overline{prop^{k^{th}}}$ and $\hat{\sigma}\left(prop_{s,n}^{k^{th}}\right)$ with $k = 1, 2, 3$, in addition to the median of the distribution. Column 6 (named "Random") presents the same statistics for the proportion of researchers who where "unlucky" and got assigned a residual workspace after they where randomized out from their 1st, 2nd, and 3rd choices. Note that the the elements from columns 3 to 6 in row "Mean" need to sum up to 1. Finally, column 7 presents one possible measure of *ex-post* efficiency: it computes the proportion of researchers who were allocated to one of their choices, relative to the proportion who got allocated to a residual workspace. A higher ratio implies that a higher proportions of researchers are getting one of their choices relative to an alternative algorithm. Importantly, such ratio weights each choice equally, which means that the ratio does not distinguish whether a researcher gets assigned to her 1st or her 3rd choice.

Table 3: Ranking revelation from 2009: Distribution of Simulated Outcomes

| | Moment | $1^{st}$ choice | $2^{nd}$ choice | $3^{rd}$ choice | Random | $\frac{1st+2nd+3rd}{Random}$ |
|---|---|---|---|---|---|---|
| | Mean | 0.635 | 0.119 | 0.037 | 0.209 | 3.77 |
| Algorithm 1 (≈ BM) | Median | 0.635 | 0.127 | 0.031 | 0.206 | 3.84 |
| | St. Dev. | (0.00) | (0.014) | (0.019) | (0.018) | |
| | Mean | 0.40 | 0.246 | 0.154 | 0.199 | 4.00 |
| Algorithm 2 (≈ DA) | Median | 0.397 | 0.238 | 0.143 | 0.206 | 3.78 |
| | St. Dev. | (0.044) | (0.044) | (0.043) | (0.027) | |
| | Mean | 0.556 | 0.181 | 0.069 | 0.194 | 4.14 |
| Algorithm 3 (RD) | Median | 0.556 | 0.175 | 0.064 | 0.190 | 4.17 |
| | St. Dev. | (0.026) | (0.035) | (0.028) | (0.021) | |

Notes: Moments computed from simulations for the preference revelation from 2009. Standard erros are in parenthesis.

There are three interesting results. First, if we guide ourselves by the measure in the last column, it seems that A1 tends to allocate a relatively high proportion of researchers to residual workspaces (i.e. workspaces that they did not include as one of their choices). The mean ratios for A1, A2, A3 are 3.77, 4.00 and 4.14 respectively. In principle, this is a negative feature of A1. However, and as we mentioned before, such ratio does not consider the "intensity" of preferences, and therefore wights with the same importance the three choices. The second observation points to the information missed by such ratio: A1 allocates a higher (on average) proportion of researchers to their first choice than A2 or A3. This is also the case if we sum the proportions allocated to their first and second choices. Third, A1 also presents a lower standard deviation for the distribution of the proportion of researchers allocated to each choice. In particular, A1 has zero variation in the proportion allocated to the first choice.

In terms of these measures, A3 fares better than A2. An interpretation of this might be that the problem of non-strategic (or "naive") choices, which A2 (an algorithm similar to a Deferred Acceptance algorithm) is supposed to correct, does not seem to be very pervasive for these preferences. As it has been pointed out by Miralles (2008) among others,

## 4.2 Robustness: Randomly Drawn Preferences

Panels in table 4 present the main results for preference drawn from a uniform and a log-normal distribution respectively. The results confirm the behavior observed for the true preferences in table 3. For both types of distribution of preferences, A1 fare worse than A" and A3 if we consider the ratio of 1st, 2nd, 3rd to random allocation. However, A1 allocates at least as many individuals to their 1st and 2nd choices as A2 and A3 (it allocates strictly more for the uniform distribution). In addition, the standard error is significantly smaller for A1, in both simulations.

Table 4: **Robustness:** Alternative preference distributions. Simulated Outcomes

**Panel A: Uniform distribution**

| | Moments | $1^{st}$ choice | $2^{nd}$ choice | $3^{rd}$ choice | Random | $\frac{1st+2nd+3rd}{Random}$ |
|---|---|---|---|---|---|---|
| Algorithm 1 (≈ BM) | Mean | 0.427 | 0.137 | 0.210 | 0.225 | 3.45 |
| | Median | 0.429 | 0.142 | 0.206 | 0.222 | 3.50 |
| | St. Dev. | (0.055) | (0.045) | (0.058) | (0.041) | |
| Algorithm 2 (≈ DA) | Mean | 0.286 | 0.218 | 0.278 | 0.218 | 3.58 |
| | Median | 0.286 | 0.222 | 0.270 | 0.222 | 3.50 |
| | St. Dev. | (0.057) | (0.051) | (0.057) | (0.043) | |
| Algorithm 3 (RD) | Mean | 0.354 | 0.180 | 0.258 | 0.208 | 3.81 |
| | Median | 0.350 | 0.175 | 0.254 | 0.206 | 3.77 |
| | St. Dev. | (0.058) | (0.050) | (0.056) | (0.040) | |

**Panel B: Log-normal distribution**

| | Moment | $1^{st}$ choice | $2^{nd}$ choice | $3^{rd}$ choice | Random | $\frac{1st+2nd+3rd}{Random}$ |
|---|---|---|---|---|---|---|
| Algorithm 1 (≈ BM) | Mean | 0.800 | 0.088 | 0.040 | 0.071 | 13.03 |
| | Median | 0.794 | 0.079 | 0.032 | 0.064 | 14.25 |
| | St. Dev. | (0.046) | (0.037) | (0.024) | (0.030) | |
| Algorithm 2 (≈ DA) | Mean | 0.653 | 0.229 | 0.078 | 0.039 | 24.86 |
| | Median | 0.651 | 0.222 | 0.079 | 0.032 | 30 |
| | St. Dev. | (0.079) | (0.056) | (0.040) | (0.027) | |
| Algorithm 3 (RD) | Mean | 0.748 | 0.14 | 0.054 | 0.057 | 16.52 |
| | Median | 0.746 | 0.143 | 0.048 | 0.048 | 19.66 |
| | St. Dev. | (0.057) | (0.046) | (0.029) | (0.027) | |

Notes: Moments computed from simulations for the preference revelation from 2009. Standard erros are in parenthesis.

# 5 Final Remarks

We have performed simulations in order to compare the performance of an algorithm implemented to allocate individuals to workspaces, with two alternative algorithms commonly used in matching problems.

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